

OLIST E_COMMERCE ANALYSIS



Prepared By
Ravichandran Sathish Kumar
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Executive Summary

- This analysis of Olist's e-commerce dataset highlights sales, customer, payment, logistics, and product performance across thousands of transactions. The dashboards uncover key business patterns and areas of improvement.
- Key Findings:
 - Sales Performance → Total sales of 15.8M across ~99k orders; strong seasonality with peaks in May and dips in September/December.
 - Customer Insights → Over 96k customers, but only 3k are repeat buyers; retention remains a major challenge.
 - Payments → Credit cards dominate (78%) with high reliance on installments (51.5% of orders).
 - Logistics → Delivery success is high (91.9% on time), but some states (Alagoas, Maranhão) suffer late deliveries above 20%.
 - Products & Sellers → Revenue is concentrated in a few top categories (Health & Beauty, Watches, Bed & Bath); top 5 categories contribute over 60% of sales.

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Overview

This project analyzes e-commerce sales data across multiple dimensions to uncover trends in customer behavior, payment patterns, delivery performance, and product performance.

The dashboards provide actionable insights to improve sales growth, customer loyalty, operational efficiency, and logistics strategy.

Dashboards Covered:

1. Sales Overview
2. Finance & Payments
3. Customer Insights
4. Logistics & Delivery Performance
5. Product & Seller Performance
6. Conclusion & Recommendations

Sales Overview

olist

Total Orders

99.44k

Average Order Value (AOV)

159.3

Total Sales

15.84M

On-time Delivery %

1

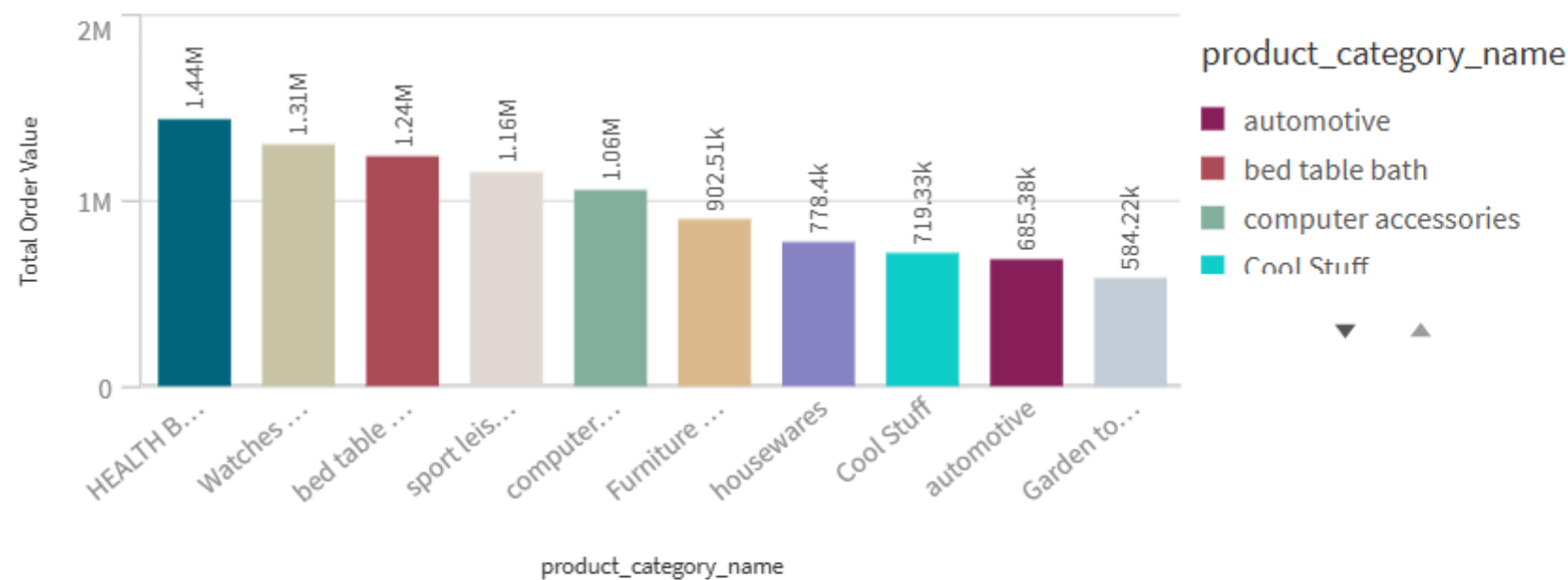
Freight % of Sales

0.14

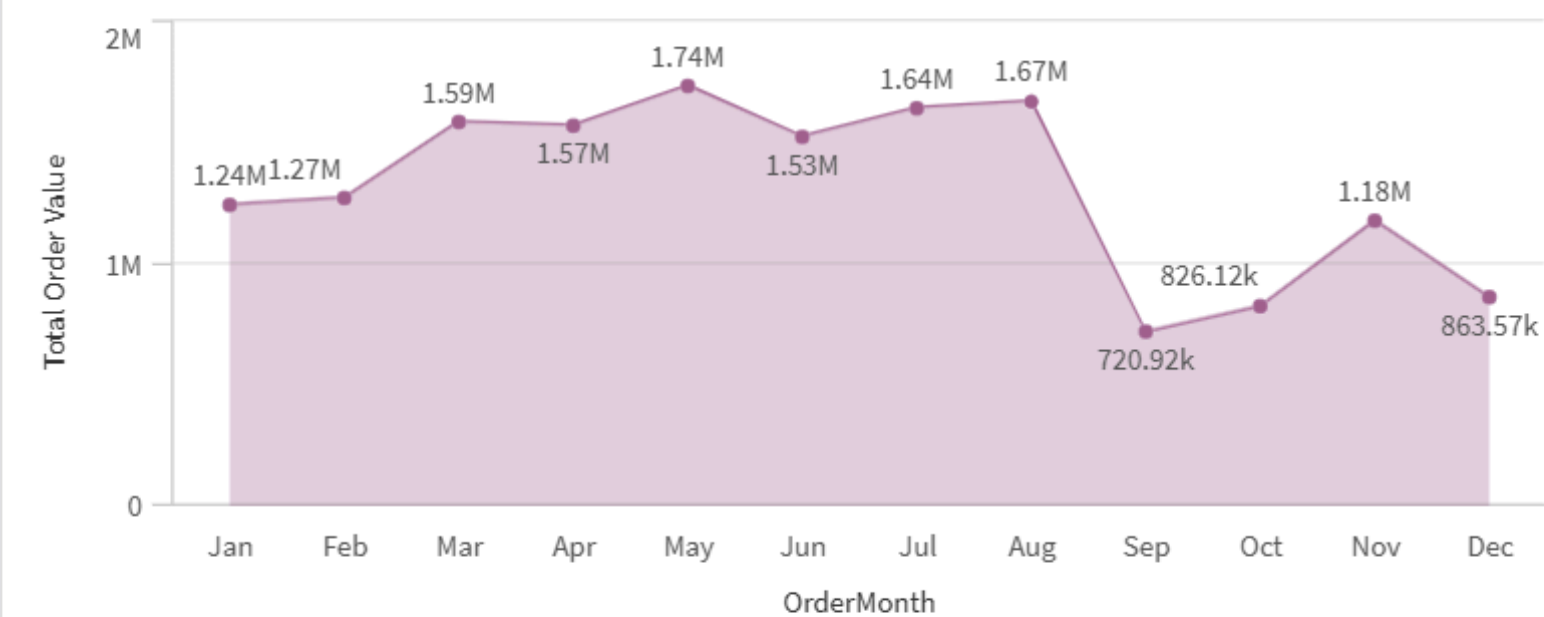
Freight Cost

2.25M

Top 10 Categories



Monthly Sales Trend



99441

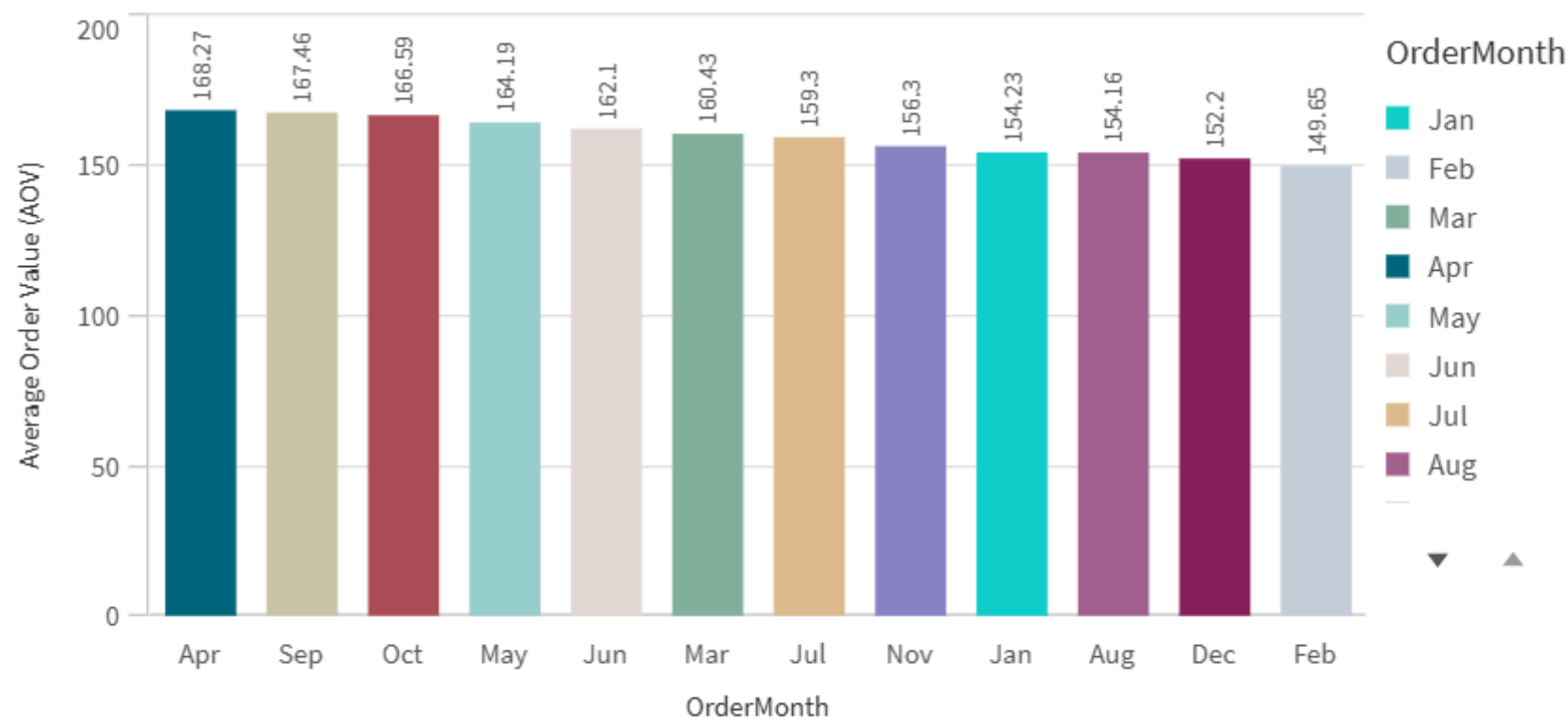
customer_city

payment_type

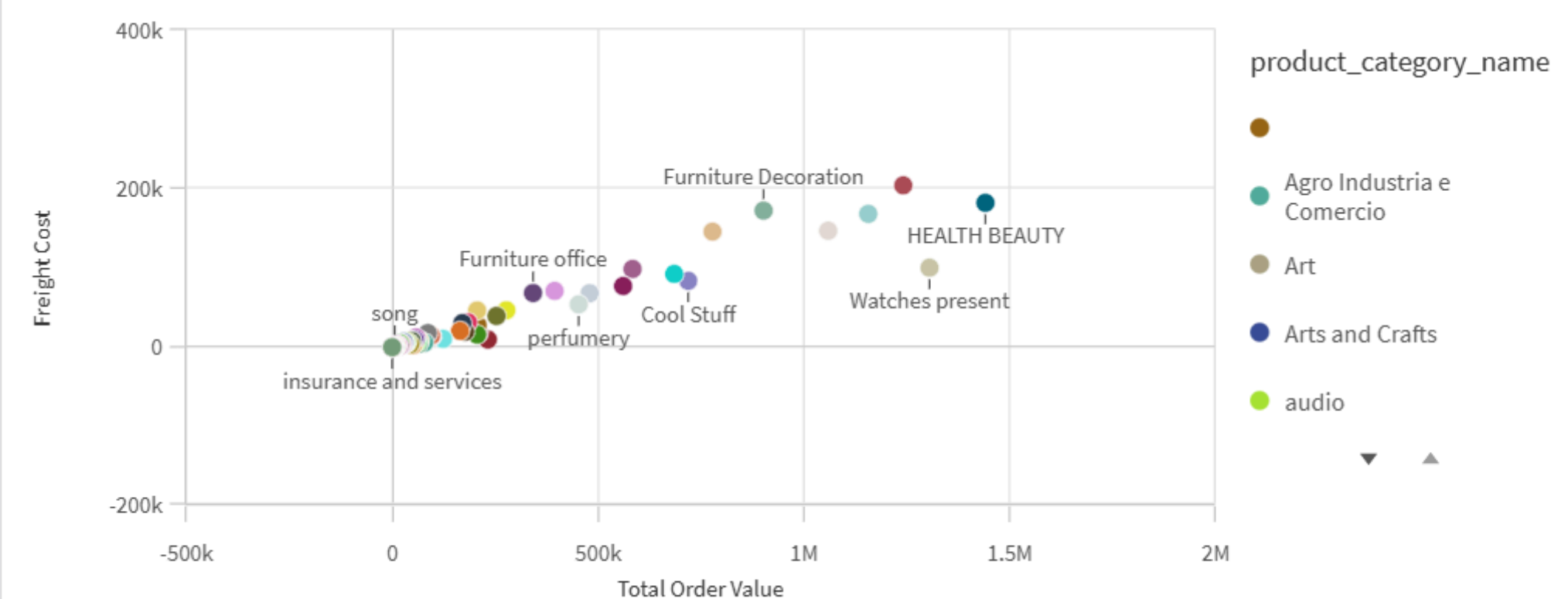
product_category_name

OrderMonthYear

Average Order Value Trend



Freight Cost vs. Sales (by Product)



Key Insights

- Total orders: 99.4k
- Total Sales: 15.8M with an Average Order Value (AOV) of 159
- Top 3 categories by order value: Health & Beauty, Watches, Bed Table Bath (together contributing over 3.8M)
- Sales peaked in May (1.74M) and were lowest in September (720k)
- Freight accounts for only 0.14% of total sales, highlighting cost-efficient shipping

Finance & Payments

Total Payment Value
16.01M

Average Payment
154.1

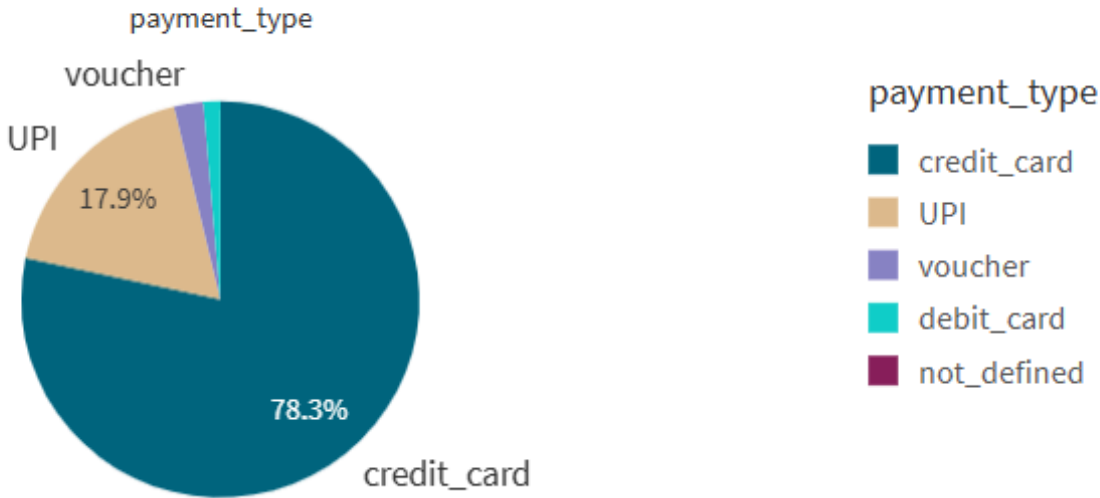
Average installments per order
2.93

Most Used Payment Method
credit_card

% Orders Paid in Installments
51.5%

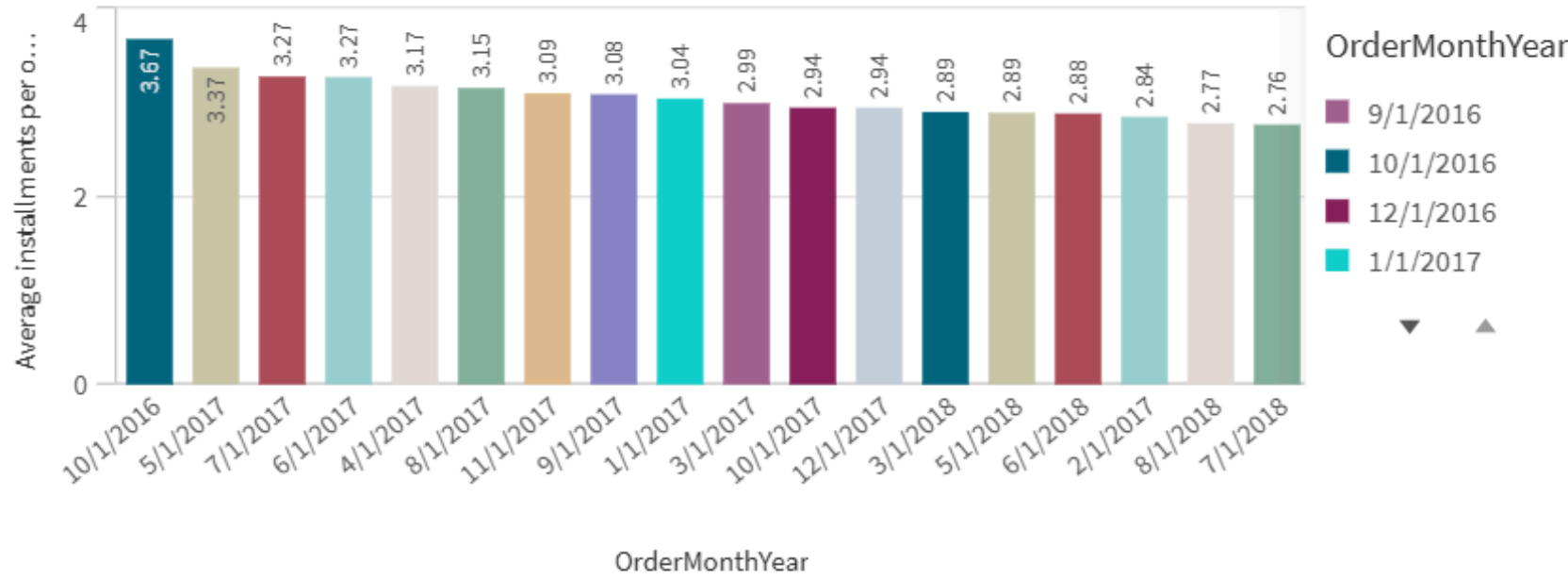
Revenue per Customer
166.6

Payment type by t



* The data set contains negative or zero values that cannot be shown in this chart.

Average installments per order



customer_state

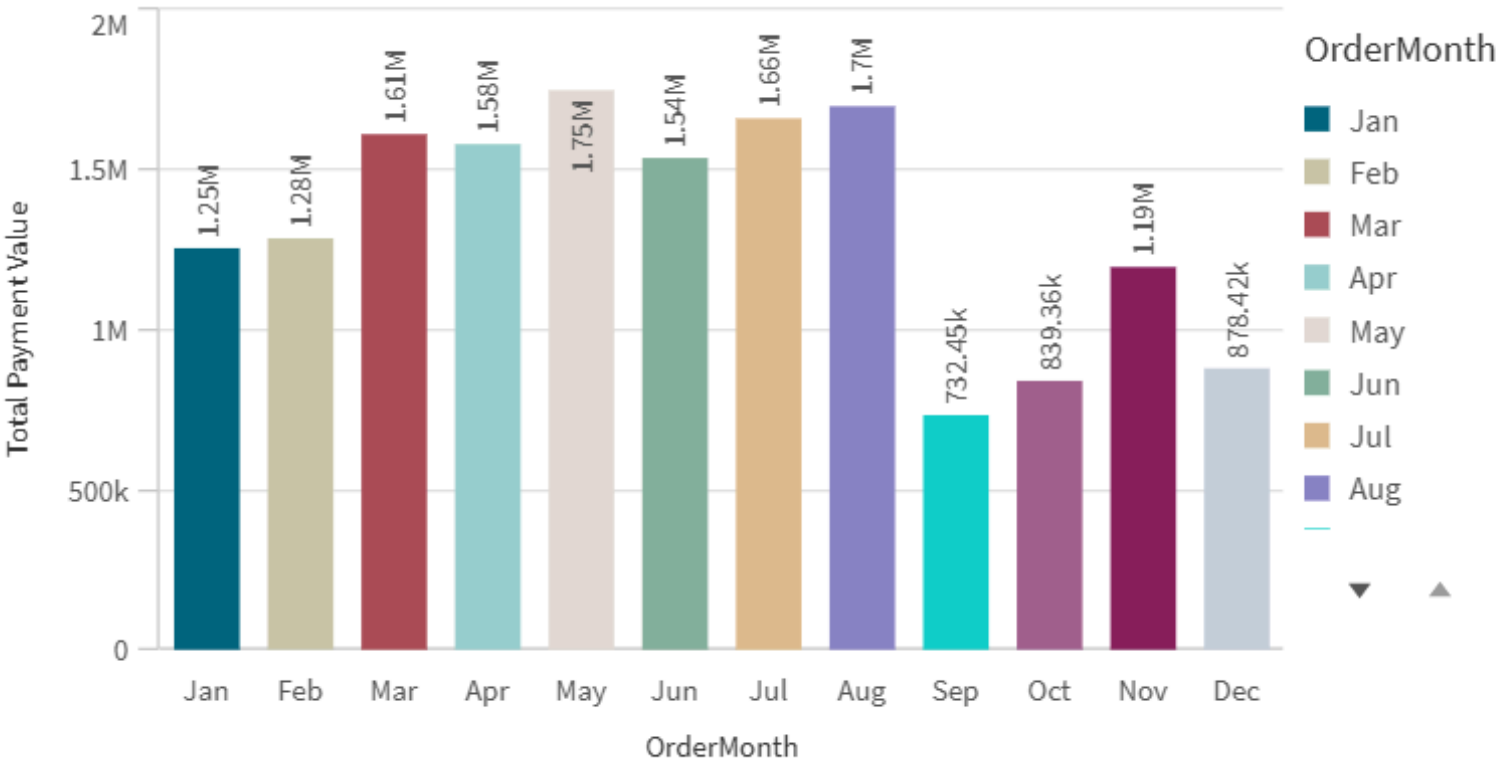
customer_city

OrderMonthYear

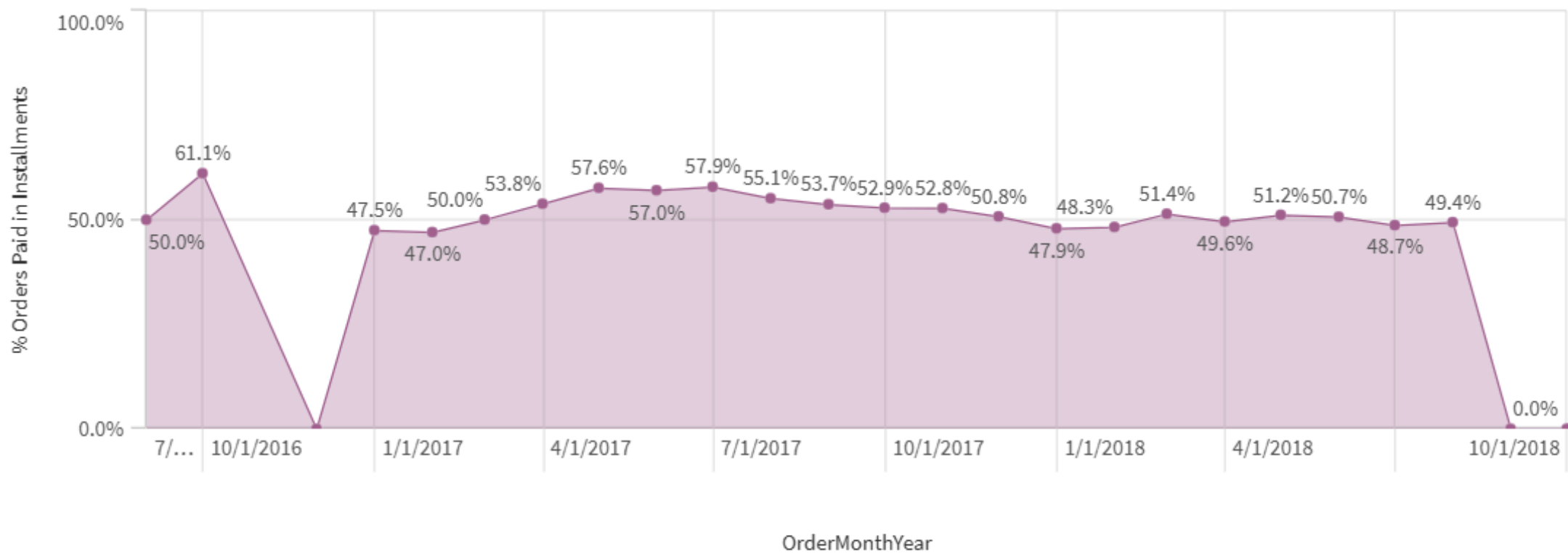
payment_type

product_category_name

Total Payment Value



% Orders paid in installments by month



Key Insights

- Total Payment Value: 16.01M
- Average Payment Value: 154
- Credit Card dominates with 78% share, followed by UPI and other modes
- 51.5% of orders were paid in installments, with an average of 2.9–3.6 installments per order
- Payment behavior is consistent across months, with installment usage showing stability

Customer Insights

Total Customers

96.1k

Average Orders per Customer

1.03

Repeat Customers

3k

Max Orders by a Single Customer

17

One Time Customers

0.97

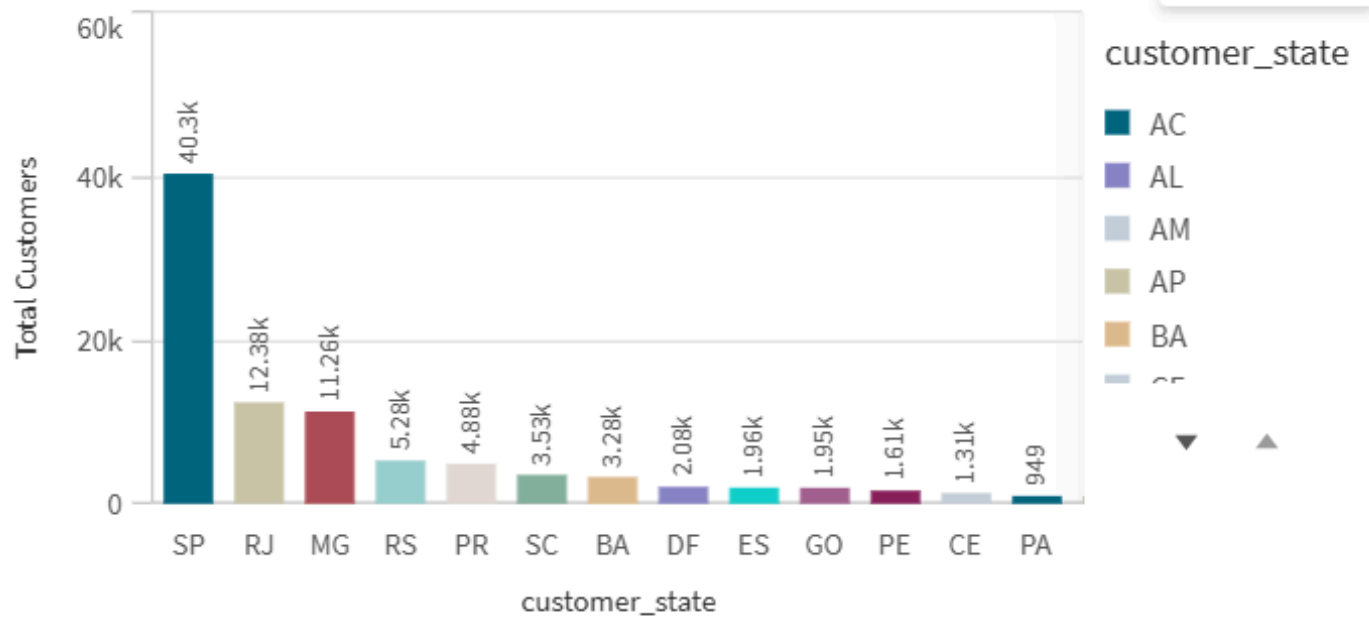
Median Order Frequency

1

New vs. Returning Customers Over Time

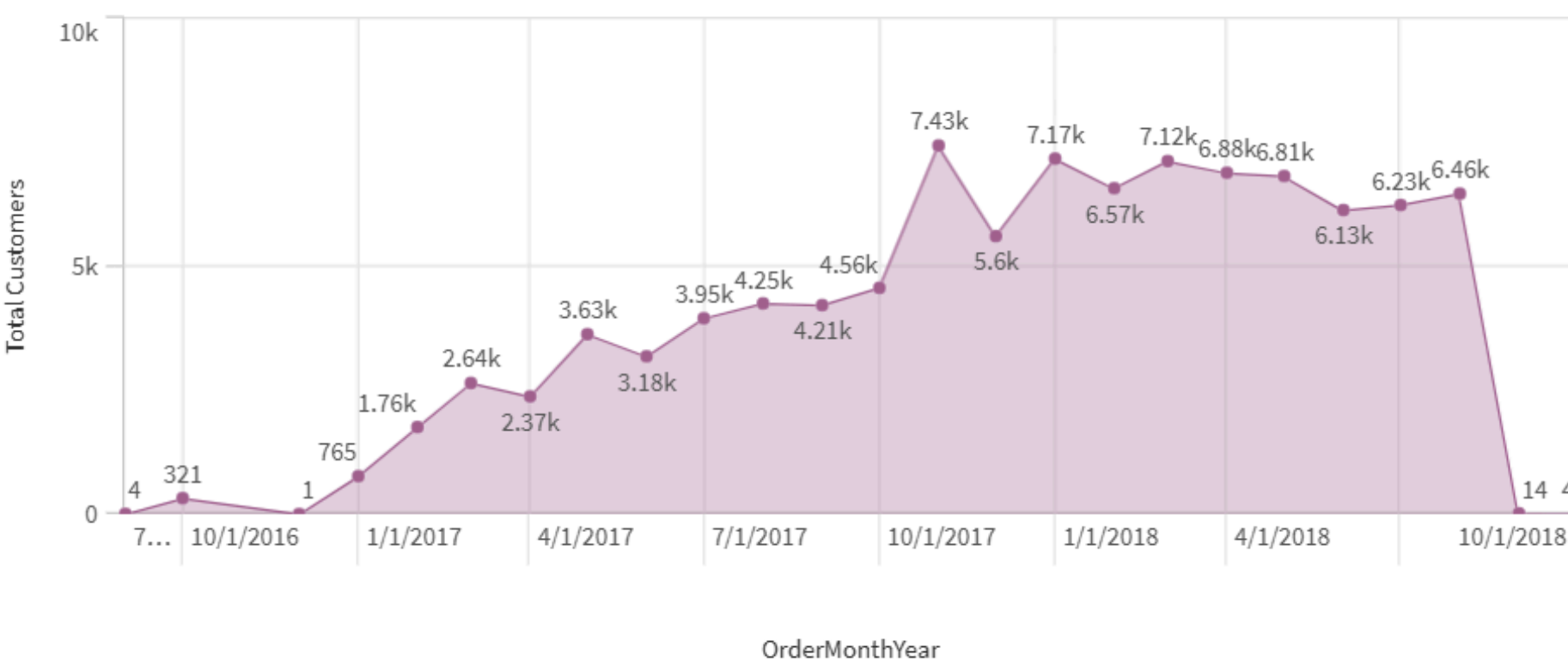


Top States by Customers

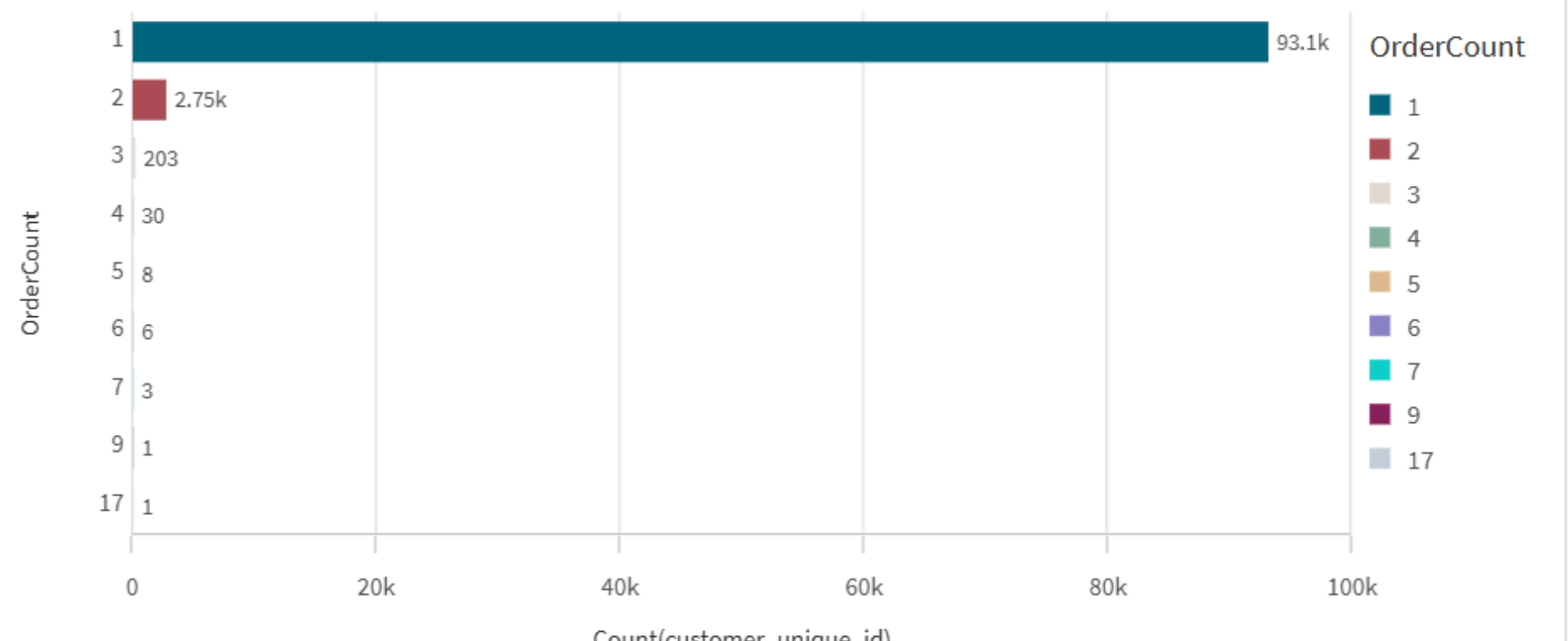


customer_city
customer_state
CustomerType
OrderYear
payment_type

Customer Growth Over Time



Customer Purchase Frequency



Key Insights

- Total Customers: 96.1k
- Only 3k repeat customers → over 93% are one-time buyers
- São Paulo is the largest customer hub with 40k customers, followed by Rio de Janeiro (12k)
- Customer base grew strongly in 2017–2018, before stabilizing
- Maximum orders placed by a single customer: 17 orders
- Customer loyalty is low → retention strategies are needed

Logistics & Delivery Performance

On-time Delivery %
91.9%

Average Delivery Time (days)
12.6 days

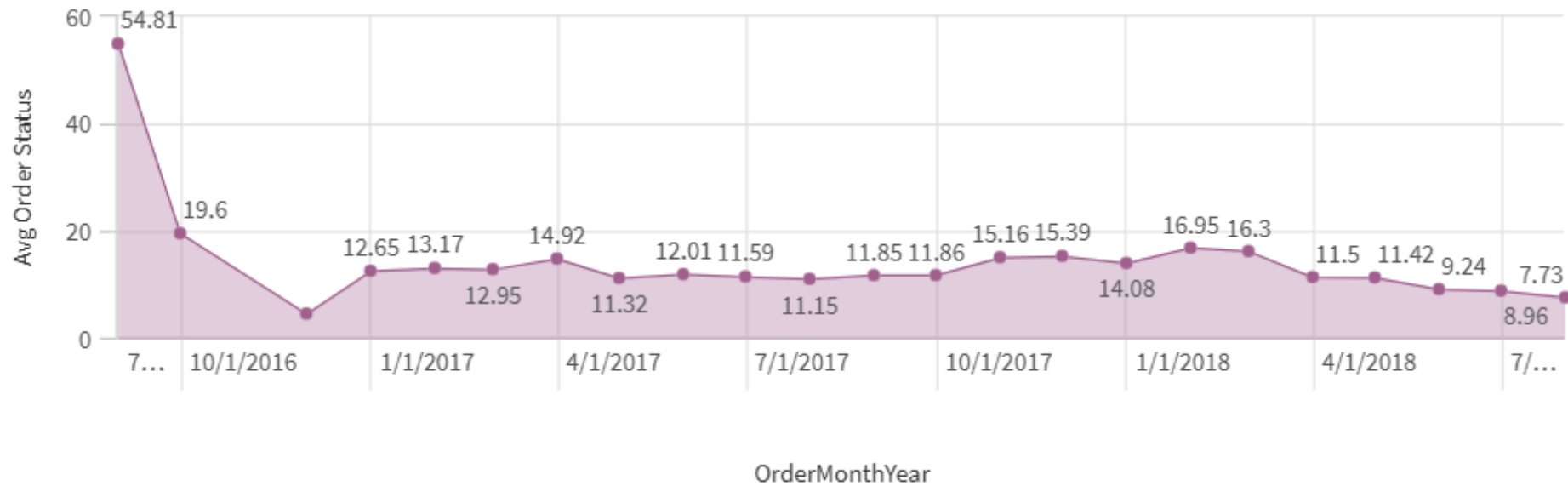
Late Delivery %
8.1%

Delivered Orders
99.44k

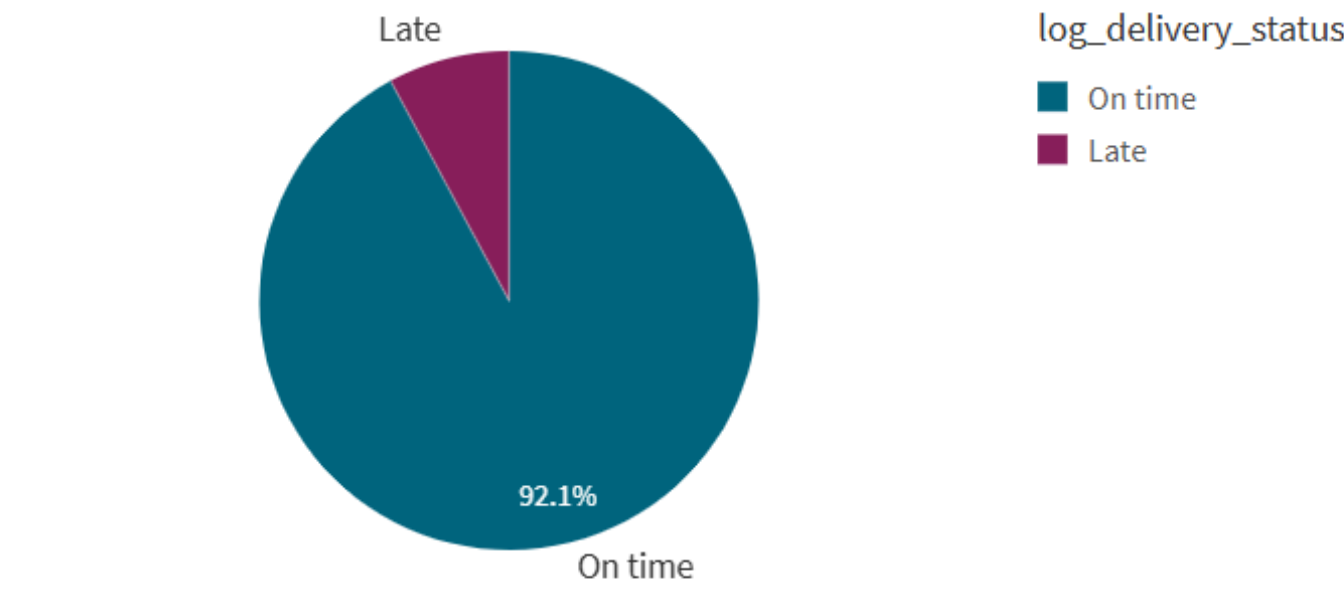
Not Delivered Orders
0

Average Delay
9.6 days

Delivery Time Trend

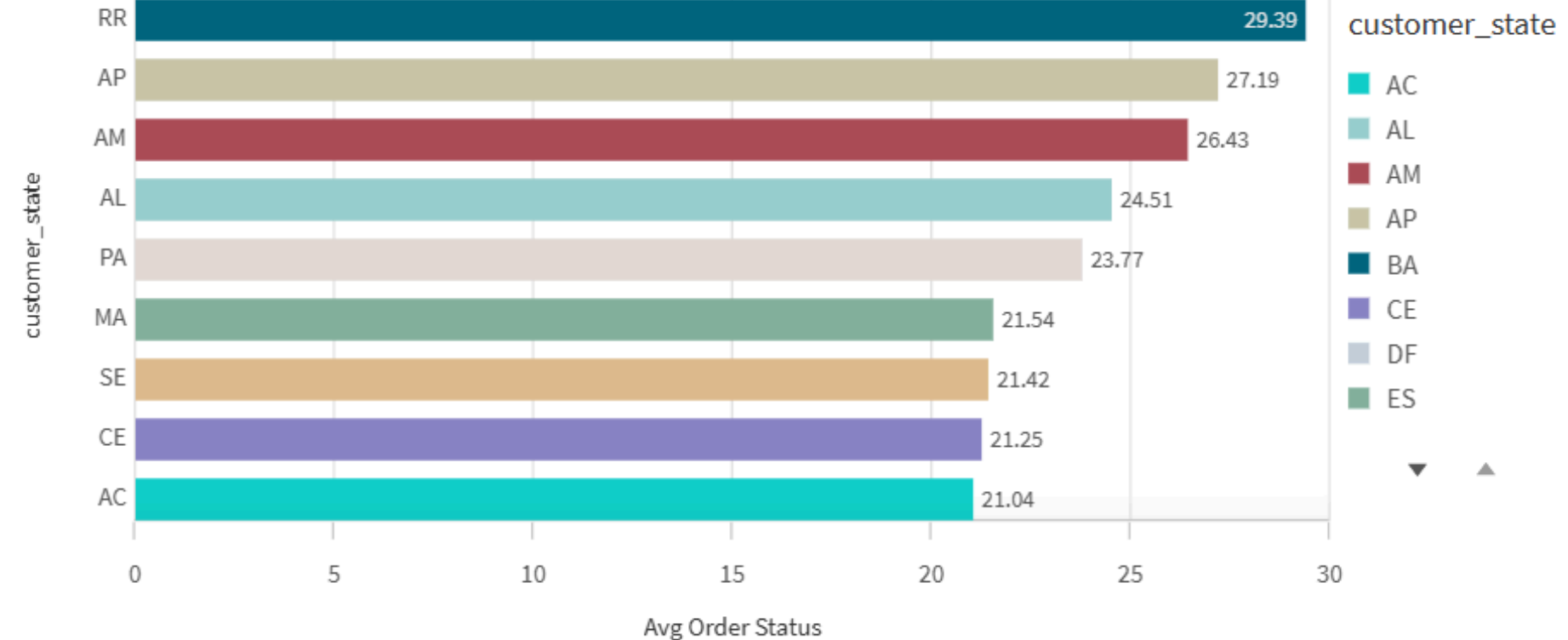


On-time vs Late Deliveries

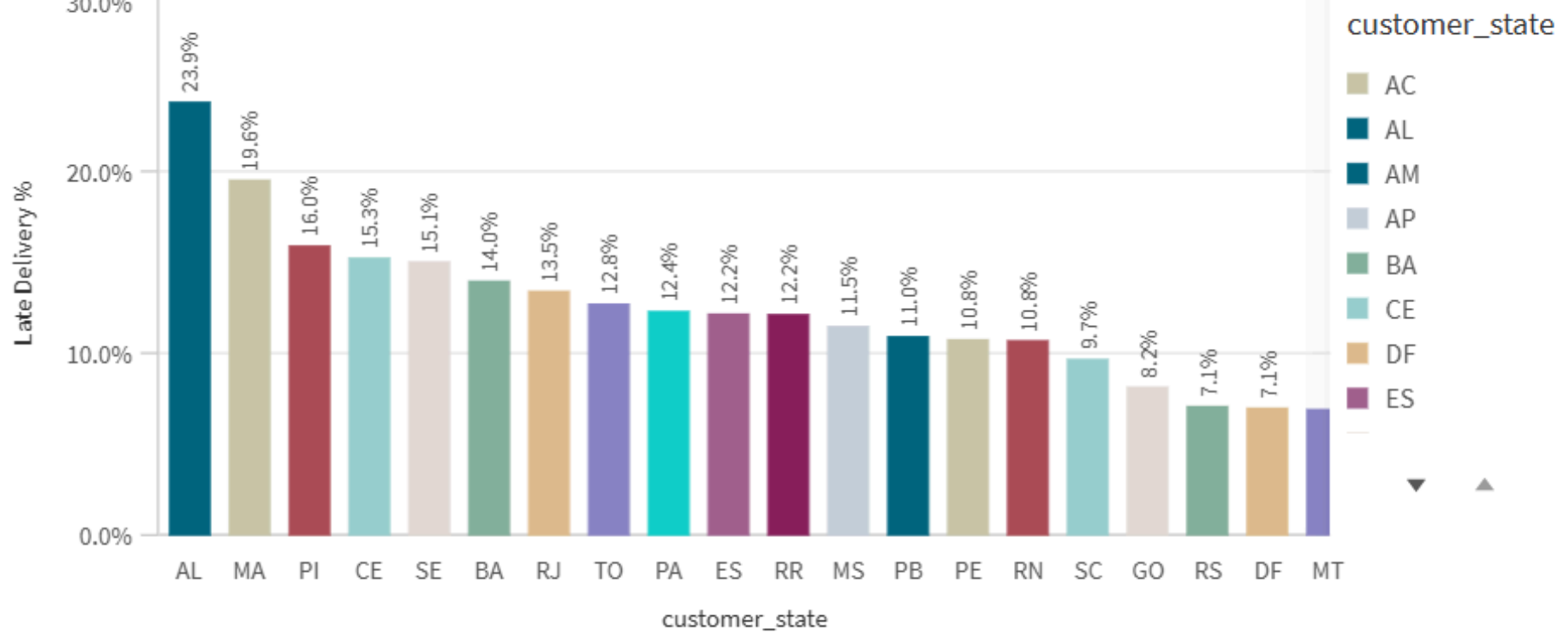


customer_state
log_delivery_status
OrderMonthYear
payment_type
product_category_name

Delivery Performance by State



Late Delivery % by State



Key Insights

- On-time Delivery Rate: 91.9%
- Late Deliveries: 8.1%
- Average Delivery Time: 12.6 days, improved to 7–9 days in recent months
- Highest late delivery states: Alagoas (24%) and Maranhão (20%)
- Logistics bottlenecks exist in specific regions → requires targeted interventions

Product & Seller Performance

Total Products Sold

32.95k

Top Product Category (by Sales Value)

HEALTH BEAUTY

Top Seller (by Sales Value)

4869f7a5dfa277a7dca6462dcf3b52b2

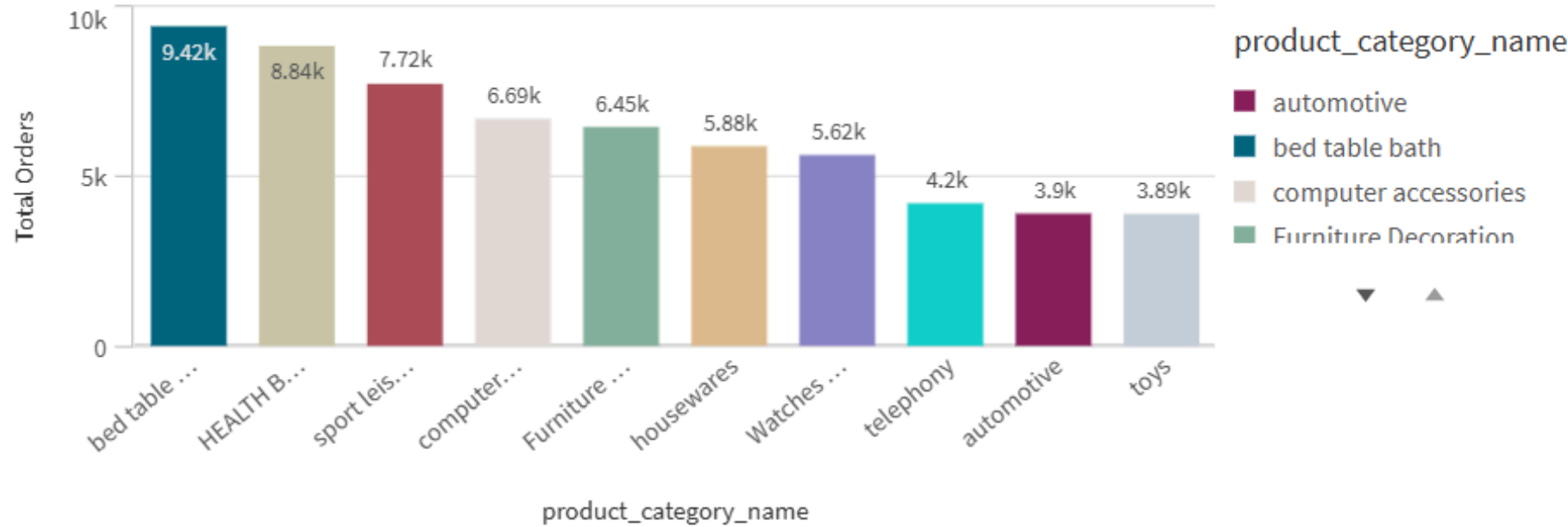
Avg Product Price

\$120.65

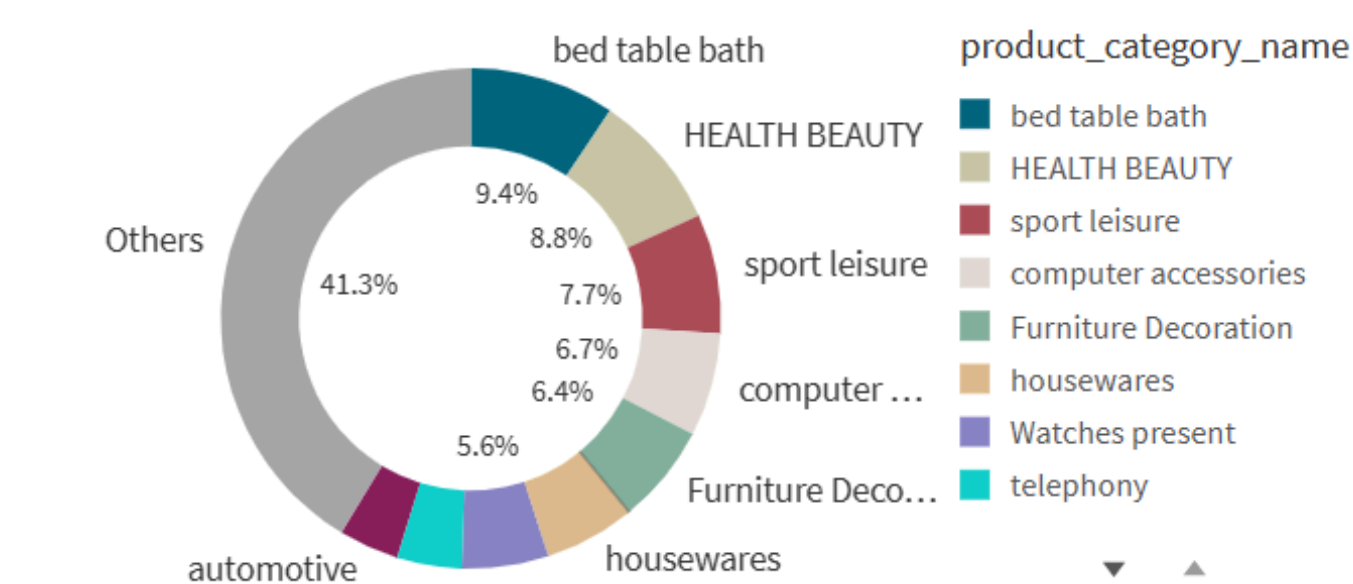
Total Revenue from Top Category

1.45M

Top 10 Product Categories by Revenue



Product Category Share



seller_city

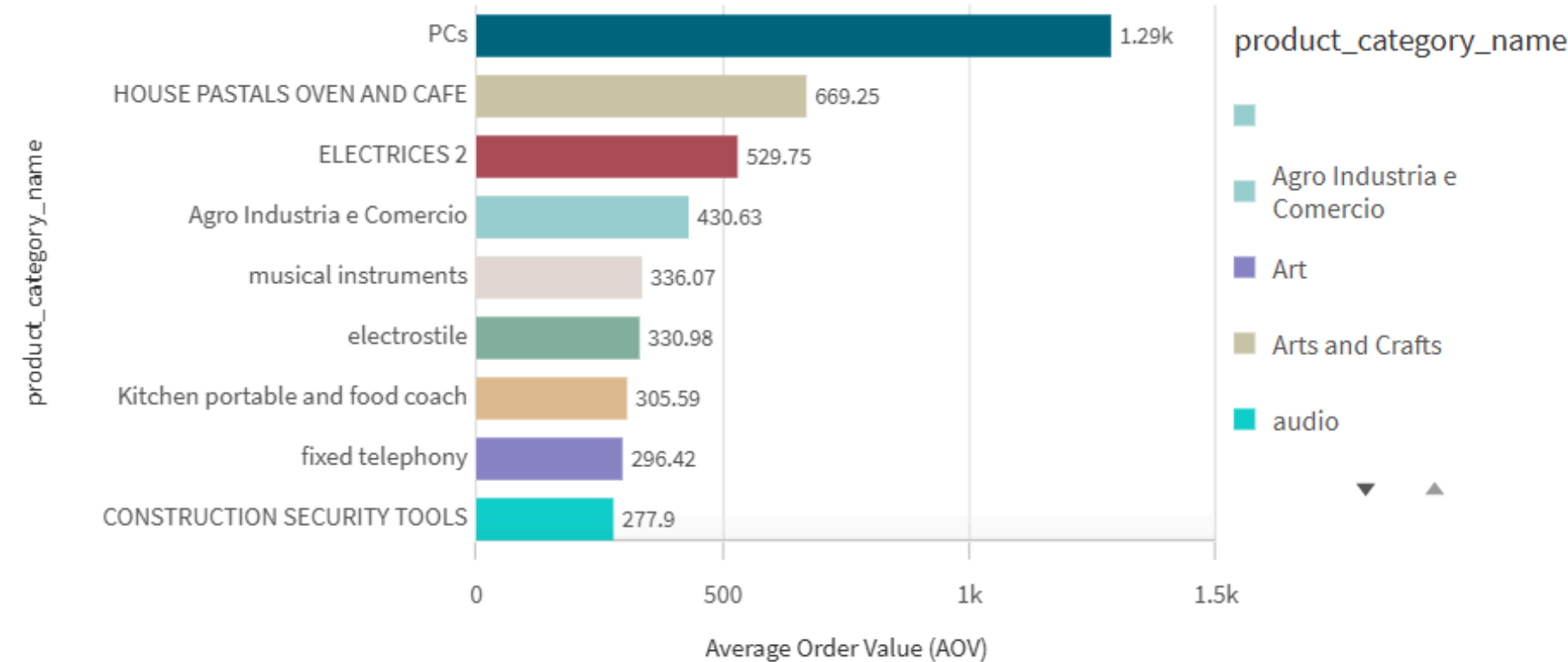
OrderMonthYear

product_category_name

payment_type

seller_state

Average Order Value by Product Category



Seller Performance by State



Key Insights

- Total Products Sold: 32.9k
- Top Product Category: Health & Beauty (1.45M revenue, 9.4k orders)
- Top 5 product categories contribute ~60% of overall revenue, indicating dependency risk
- High AOV products: PCs (\$1.29k), House Pastals Oven & Café (\$669), Electricres (\$530)
- Sellers are globally spread, but sales concentration is visible among a few strong categories and states

Conclusion

The e-commerce business demonstrates strong sales volume and broad product coverage, but there are critical challenges:

- Revenue concentration in a few product categories and sellers
- Low customer loyalty, with most being one-time buyers
- Regional delivery inefficiencies, particularly in certain states
- Heavy reliance on credit card payments and installments

Recommendations

1. Boost Customer Retention

- Launch loyalty programs, cashback offers, and personalized promotions
- Encourage repeat purchases with bundled deals or referral benefits

2. Optimize Logistics Performance

- Strengthen last-mile delivery in late-delivery states (Alagoas, Maranhão)
- Partner with local couriers for faster regional coverage

3. Leverage Payment Insights

- Promote larger basket sizes by highlighting installment flexibility
- Introduce promotional financing options for premium categories

4. Diversify Product Portfolio

- Reduce dependency on top 5 categories by investing in underperforming but emerging categories
- Support niche sellers to expand variety and attract new customers

5. Seasonal Campaigns

- Address dips in sales months (e.g., September, December) with discounts, flash sales, or targeted campaigns

<https://github.com/sathishgrm/Data-Analytics>